

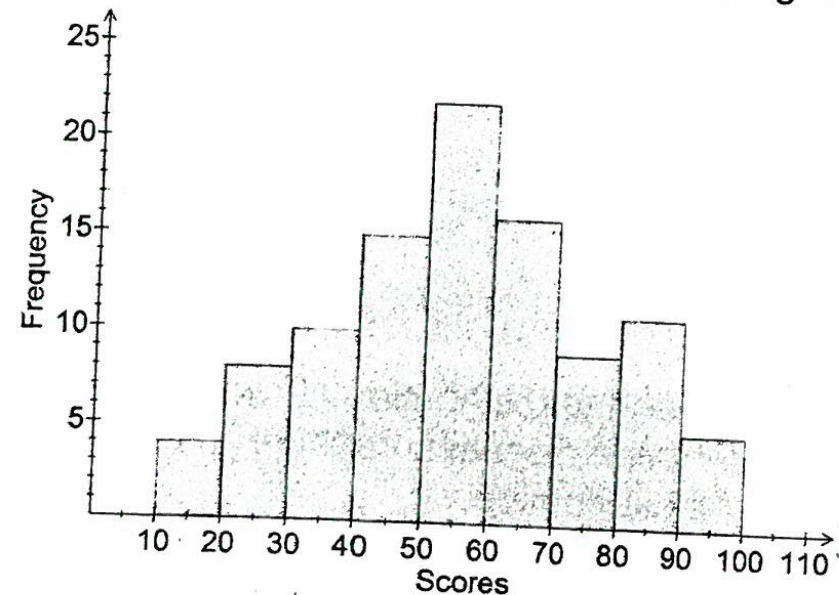
Interpreting Histogram

Describe the distribution of the scores displayed by the histogram below under the given headings.

(a) Measures of spread:

(b) Measures of central tendency:

(c) Shape:



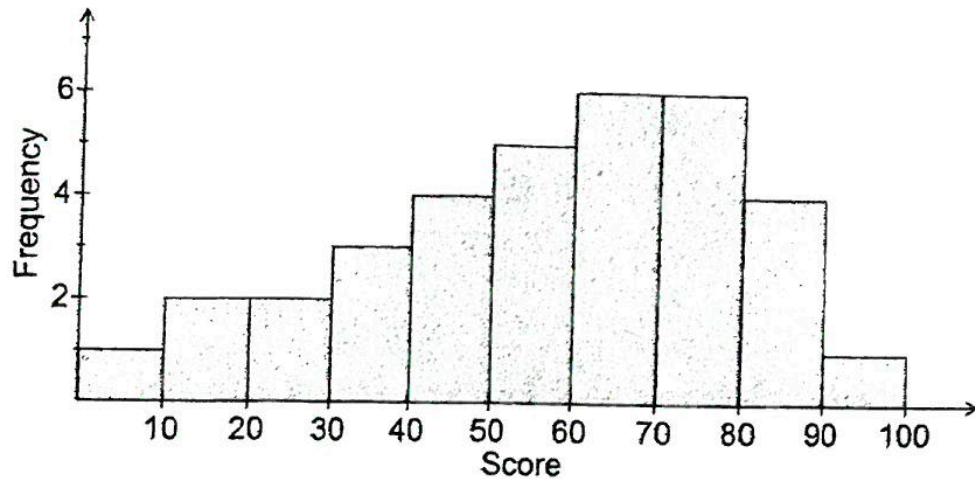
1. (a) The data displayed by the histogram involves 100 scores. The scores are spread between about 10 and 100 with a range of about 90. The standard deviation has been estimated to be 20.51 (2d.p.) using the class midpoints

(b) The mean has been estimated to be 56.2 by using the class midpoints. As the total number of scores is 100 the ~~middle~~ mean score is in the range 50 to 60, that is the median class of the scores is 50 to 60 marks. The modal class is the same as the median class that is 50 to 60.

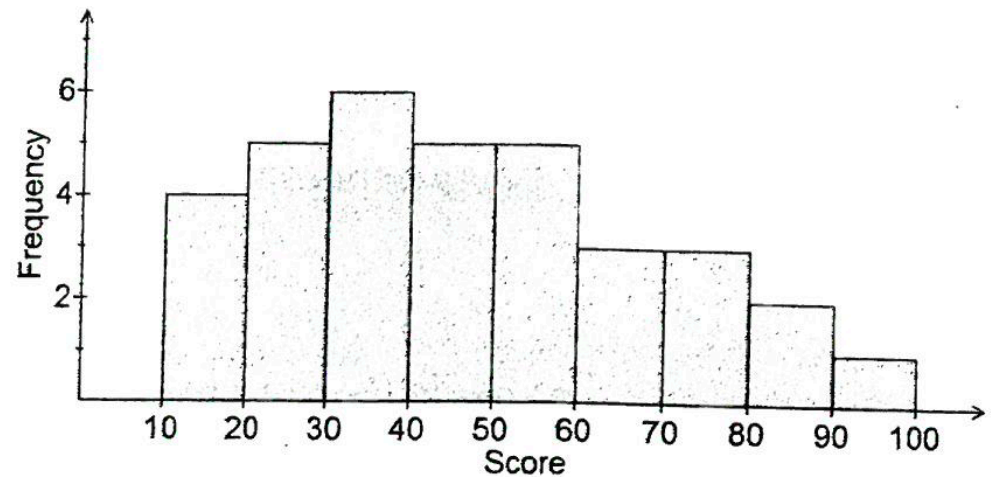
(c) The distribution of scores is unimodal, symmetrical, peaking at the median/modal class of 50 to 60 and falling away reasonably symmetrically left and right of this central column.

1. Consider the following data sets:

Data Set A



Data Set B



- (a) Estimate the mean and standard deviation for both data sets.
- (b) Describe the shape of the data for both data sets.
- (c) Comment on the similarities and differences between data sets A and B.

1. (a) Mean A = 56.47 (2 d.p.) Std dev'n A = 22.51 (2 d.p.)

Mean B = 46.47 (2 d.p.) Std dev'n A = 21.85 (2 d.p.)

(b) Data set A: ~~The distribution is unimodal over the interval 60 to 80 with just over 95% of scores in this group.~~ The scores

The distribution is bimodal
and the modal classes are
60 to 70 and 70 to 80

are clustered about 70 and their distribution is negatively skewed

Data set B: The distribution is unimodal over the interval 30 to 40. The scores are clustered about the modal score interval with approx. 74% of the scores in this cluster between 10 and 60.. The distribution of the scores for this data set is positively skewed.

(c) Both data sets have 34 data points.

The mean for data set A is significantly greater than the mean for data set B, 56.47 compared to 46.47 indicating that the scores for set A are much higher than those for set B.

The scores for data set A are slightly more spread than those of data set B as indicated by standard deviations of 22.51 and 21.85 respectively and the observation that the scores for A range from 0 to 100 and those for B range from 10 to 100.

Both distributions are skewed, distribution A is skewed to the left and distribution B is skewed to the right.